

Seth DeWhitt

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EDUCATION

Purdue University, West Lafayette, IN

August 2023 – May 2027

B.S. in Computer Science, conc. Software Engineering & Security

GPA: 3.6/4.0

Relevant Coursework: Data Structures, Analysis of Algorithms, Software Engineering, Systems Programming, Computer Architecture, C Programming, Software Testing, Object-Oriented Programming, Linear Algebra

Activities: Jazz Education Network (President), Purdue Jazz Band (Lead Tenor Sax.)

Awards: Elmhurst U. Jazz Fest Outstanding Soloist (4x), Dean's List (3x), H.M. StarkHacks 2026 (AMD Track)

EXPERIENCE

Research Lead & Teaching Assistant

August 2025 – Present

The Data Mine, Purdue University | Agile, Python, TypeScript, Surreal, Firebase

West Lafayette, IN

- Directing 11 student researchers in collaboration with the Practical AI Podcast (**1.5M+ listens**)
- Implementing RAG chatbot trained on **349+ episodes** with citation-backed responses
- Designed scalable podcast generation platform integrating LLM-produced transcripts and voice clone models

Software Engineer Intern

June 2025 – August 2025

Chewy | Python, React, Vite, FastAPI, Docker, AWS, Terraform, Jenkins, PostgreSQL

Bellevue, WA

- Deployed scalable & containerized development environments with custom tooling for in-house model execution on AWS ECS/Fargate, **reducing setup overhead by 80%** and supporting **15+ engineers**
- Automated distributed system maintenance with AWS Lambda and Kinesis, **lowering AWS expenses by 25%**
- Engineered **persistent storage architecture** linking Okta SSO authentication with AWS EFS, ECS, and RDS to support secure stateful development sessions across container restarts
- Designed and implemented Terraform-based infrastructure provisioning and Jenkins CI/CD pipelines, **decreasing deployment times by 66%**

Undergraduate Data Science Researcher, Bayer Crop Sciences, Syngenta

August 2023 – May 2025

The Data Mine, Purdue University | Python, R, Sci-kit learn, Pandas

West Lafayette, IN

- Trained a LightGBM model on **500k+ rows** to model climate change impact on corn yields at **91% accuracy**
- Optimized model performance through feature engineering and cross-validation, reducing RMSE by **24%**
- Implemented data preprocessing and QC pipelines for **100M+ data points**, validated using mixed linear models

PROJECTS

ControVirtual | Python, C#, Flask, PyTorch, OpenCV, LeRobot | GitHub | DevPost

April 2026

- Streamlined and upscaled use of AMD SO-101 Robot Arms for industry experts via remote use of a Meta Quest S3, voice & natural language processing, and a Vision-Language-Action (VLA) model
- Enabled spatial visibility of remote arms by livestreaming a camera feed as well as displaying 3D replica arms
- Honorable Mention** for "Best Use of AMD Tech" at StarkHacks 2026

SlopMop | Python, React, Vite, Firebase, AWS, PyTorch | GitHub

January 2026 – April 2026

- Engineering Chrome/Firefox extension that auto detects AI social media content, leading a team of 6
- Achieved **>90% code coverage** testing for extension + website with Vitest and Jest
- Deployed a hybrid system of image detection models, maximizing accuracy & lowering response times to **<3s**

Evallm | TypeScript, React, Next.js, Vercel, OAuth2, MongoDB | GitHub | Link

July 2025 – August 2025

- Architected and deployed full-stack platform for **benchmarking LLM outputs**, integrating popular LLMs with experiment tracking and analytics pipelines
- Integrated **real-time model inference**, enabling low-latency feedback and interactive evaluation workflows

UNIX-like Shell | C++, Lex/Yacc

February 2025 – April 2025

- Implemented POSIX-style shell in C++ with advanced parsing (Lex/Yacc), supporting **I/O redirection, pipelines, subshell execution, and signal handling**
- Handled process management using fork/exec & pipe syscalls, enabling **concurrent processes** and zombie reaping
- Constructed command history and enabled wildcard expansion with **O(n) lookup**

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, R

Frameworks & Runtimes: React, Next.js, Node.js, Flask, FastAPI, Tailwind

Databases & Tools: PostgreSQL, MongoDB, AWS, Docker, Terraform, Git, Jenkins, Pandas, Flyway, Vite